AI-Driven Chronic Pain Management System

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Abstract

Chronic pain affects millions worldwide and relies on subjective reporting.

This project uses wearable physiological signals to classify pain levels (1–8).

Machine learning model helps to improves accuracy and provides real-time monitoring.

Introduction

Chronic pain lasts longer than 3 months and impacts physical, emotional, and mental health.

Wearable sensors capture measurable physiological changes related to pain.

This project develops a machine-learning-based system to classify pain levels using **physiological signals collected from a watch device**.

Existing System

Pain assessed using:

i. Numerical Rating Scale (1–8)
ii. Visual Analogue Scale
iii. Manual doctor examination
iv. Completely dependent on patient feedback
v. Not continuous, not automated

Limitations of Existing Systems

Highly subjective

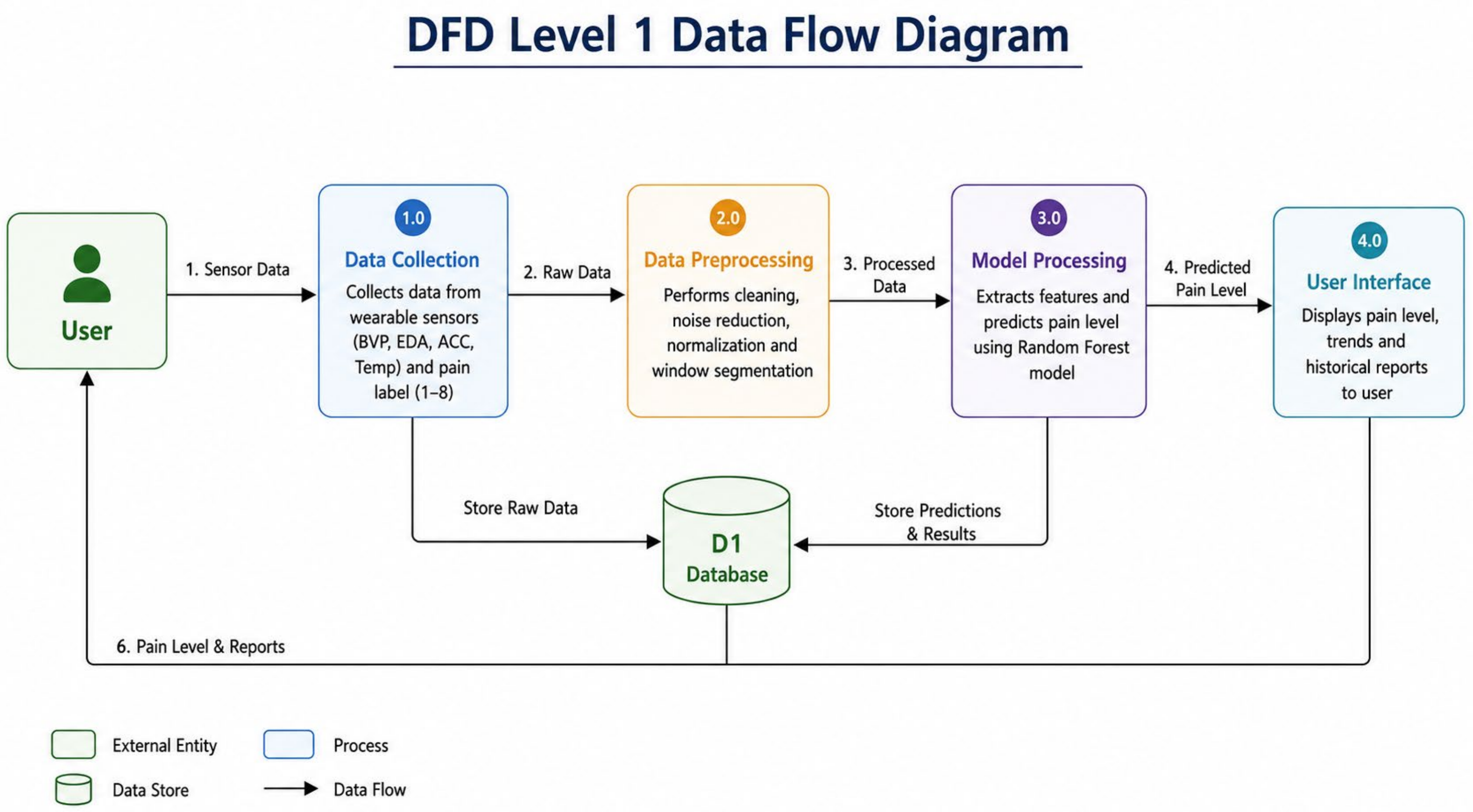
i. Inaccurate for patients who cannot express pain
ii. Not real-time
iii. Do not capture physiological reactions
iv. No automated predictions

Proposed System

An AI-based system that uses *wearable physiological signals* to predict pain intensity.

Features:

- Provides instant real-time monitoring.
- Measures pain using non-invasive sensors.
- Allows continuous wearable pain tracking.
- Predicts pain level on 1–8 scale.


DFD Level 1 Data Flow Diagram
User
1. Sensor Data
1.0
Data Collection
Collects data from wearable sensors (BVP, EDA, ACC, Temp) and pain label (1–8)
2. Raw Data
2.0
Data Preprocessing
Performs cleaning, noise reduction, normalization and window segmentation
3. Processed Data
3.0
Model Processing
Extracts features and predicts pain level using Random Forest model
4. Predicted Pain Level
4.0
User Interface
Displays pain level, trends and historical reports to user
Store Raw Data
D1
Database
Store Predictions & Results
6. Pain Level & Reports
External Entity
Process
Data Store
Data Flow

Methodology

1. Data Collection

- Physiological data collected using wearable biosensors
- Parameters used:

  i. Heart Rate (HR)
  ii. Electrodermal Activity (EDA)
  iii. Temperature
  iv. Blood Volume Pulse (BVP)

- Dataset obtained from Kaggle for chronic pain prediction analysis

2. Data Preprocessing

- Removing missing values
- Cleaning noisy sensor data

3. Feature Extraction

- Extraction of important physiological features from sensor signals.
- Features extracted:

  i. Heart Rate (HR)
  ii. Electrodermal Activity (EDA)
  iii. Blood Volume Pulse (BVP)
  iv. Body Temperature

4. Model Training

- Split into train/test sets.
- Models trained: Random Forest, SVM, Gradient Boosting.

5. Model Evaluation

- Accuracy, F1-score, precision, recall.
- Confusion matrix for all 5 pain levels.
- Comparison between ML models.

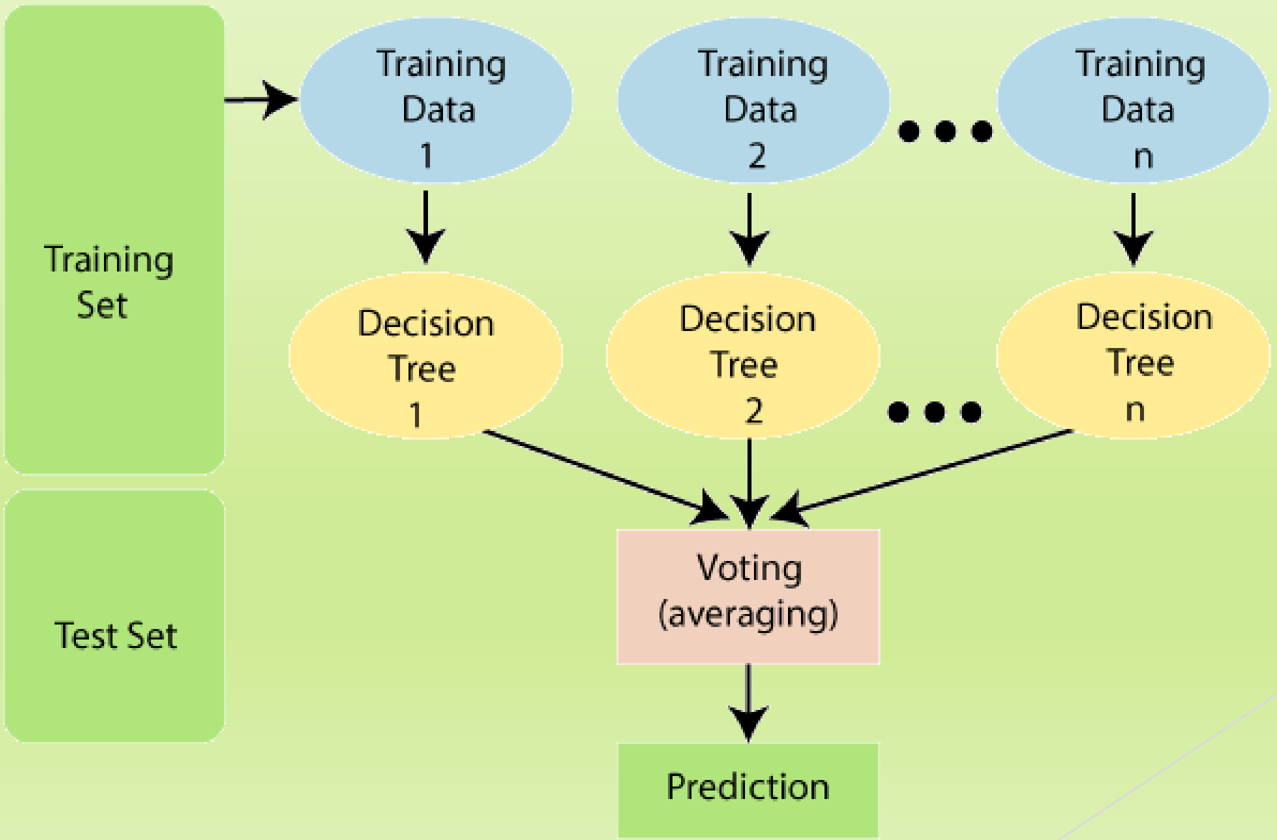
6. Prediction

- Final model outputs **Pain Level (1–8)**.
- Real-time compatibility for wearable systems.

Machine Learning Algorithms

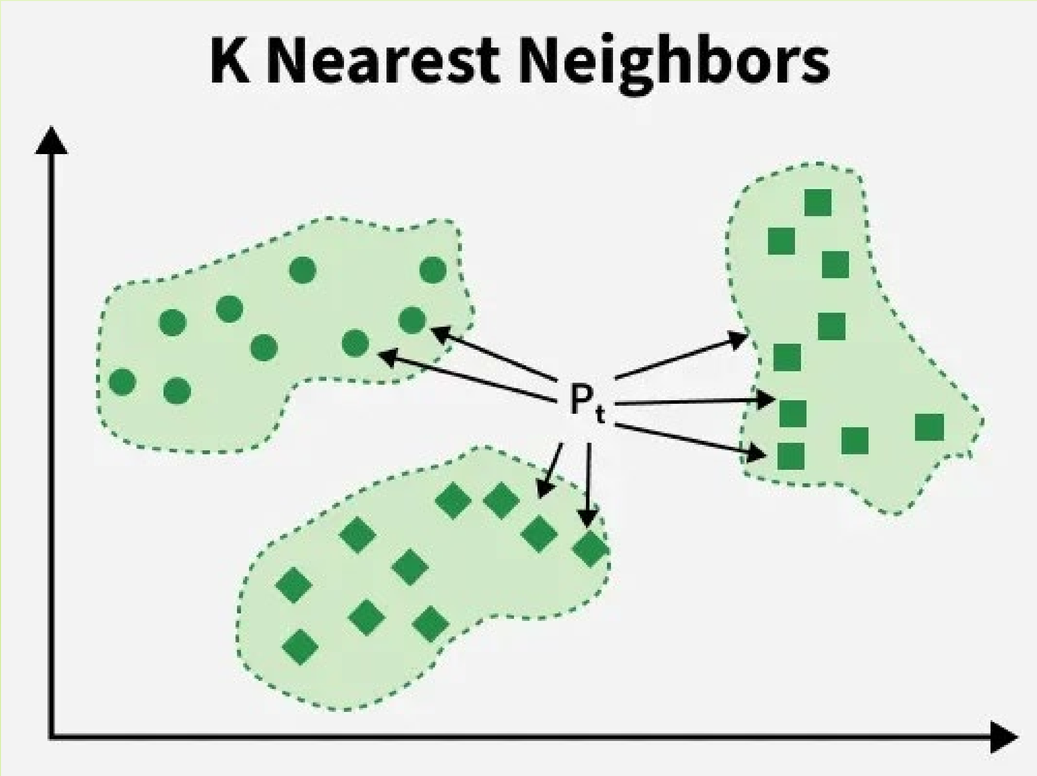
Random Forest:

- Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique. It can be used for both Classification and Regression problems in ML. It is based on the concept of **ensemble learning,** which is a process of *combining multiple classifiers to solve a complex problem and to improve the performance of the model.*

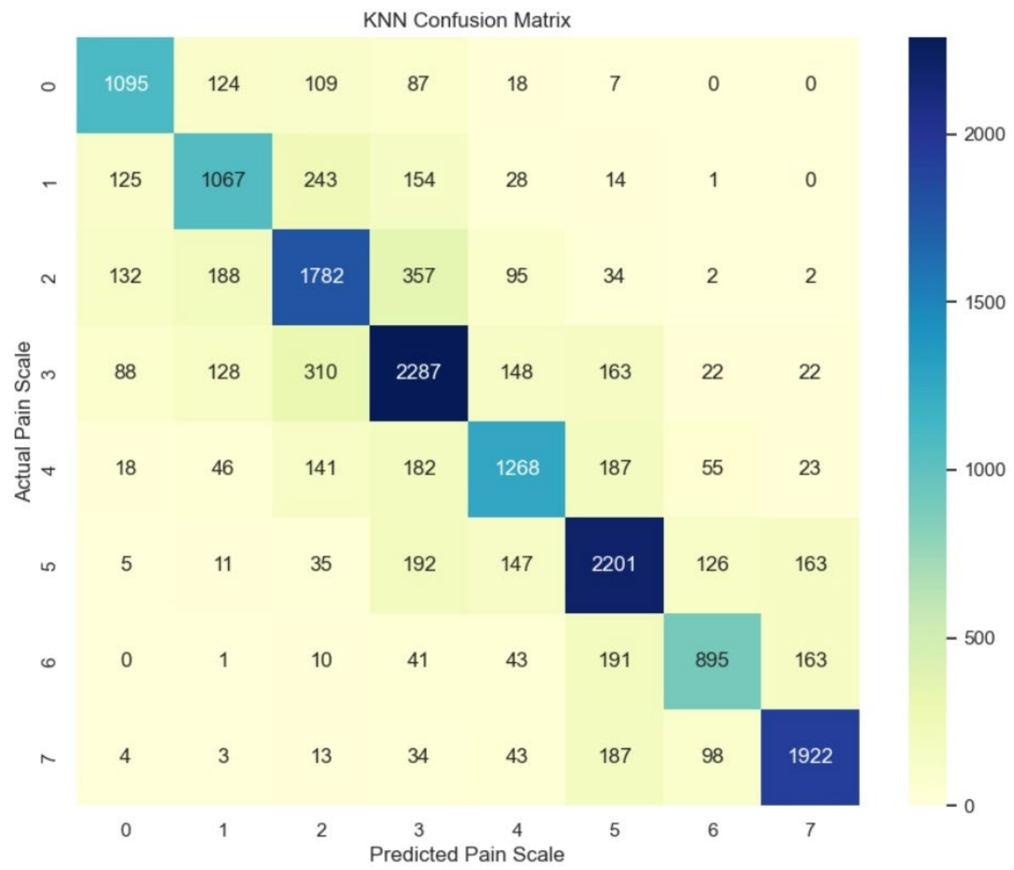


K-Nearest Neighbor(KNN)

- KNN is one of the most basic yet essential classification algorithms in machine learning. It belongs to the supervised learning domain and finds intense application in pattern recognition, data mining, and intrusion detection.
- It is widely disposable in real-life scenarios since it is non-parametric, meaning it does not make any underlying assumptions about the distribution of data (as opposed to other algorithms such as GMM, which assume a Gaussian distribution of the given data). We are given some prior data (also called training data), which classifies coordinates into groups identified by an attribute



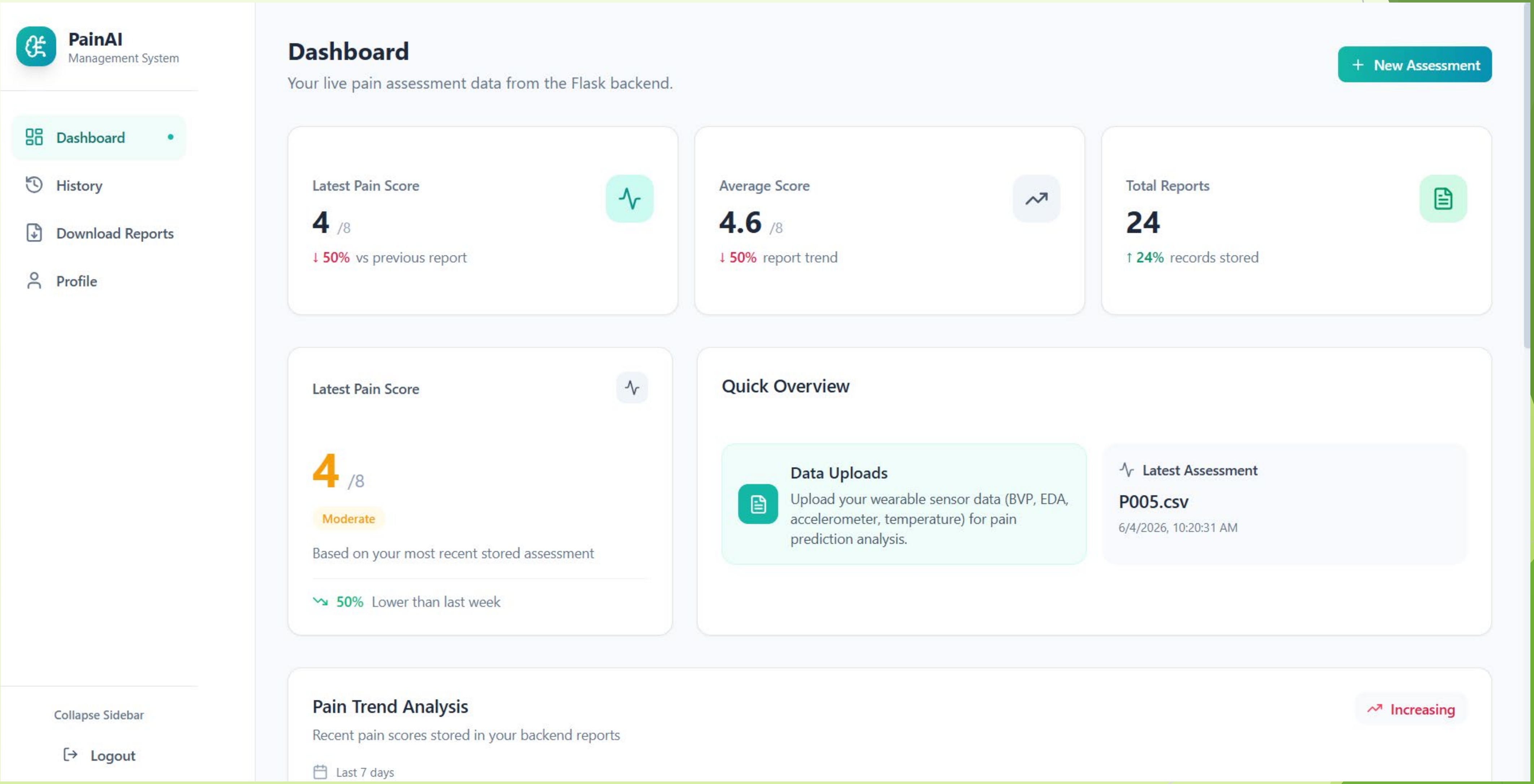
Accuracy of Different Models



Random Forest
Accuracy-79%

KNN
Accuracy-72%

Sample Output:



Conclusion

- The AI-driven chronic pain management system accurately predicts pain levels using wearable sensor data.
- It offers objective, continuous, and automated pain assessment, reducing dependency on subjective reporting and improving healthcare decision-making.

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